

Course: B. Tech. Branch : Artificial Intelligence & Data Science Semester :III

Subject Code & Name: BTAIC302 & An Introduction to Artificial Intelligence

Max Marks: 60

Date:

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.		12
A) Define artificial intelligence and elaborate the applications of artificial Intelligence in the real world.	CO1	6
B) Explain History of Artificial Intelligence.	CO1	6
C) List types of AI agent and explain any one.	CO1	6
Q.2 Solve Any Two of the following.		12
A) What do you mean by Generate and Test	CO2	6
B) Give and Explain Hill Climbing algorithm.	CO2	6
C) Illustrate Beam search algorithm?	CO2	6
Q. 3 Solve Any Two of the following.		12
A) Differentiate between Uninformed and Informed Search technique.	CO3	6
B) Compare and explain Depth First search and Breadth First search methods.	CO3	6
C) Explain iterative deepening depth first search (IDDFS) and justify its parameters based on time complexity	CO3	6
Q.4 Solve Any Two of the following.		12
A) With an example explain A*algorithm. State the properties of A* algorithm.	CO3	6
B) Explain the iterative deepening A* algorithm.	CO3	6
C) Explain importance of pruning CLOSED and OPEN list of A* algorithm.	CO3	6
Q. 5 Solve Any Two of the following.		12
A) Give Algorithm for Backtracking and Look ahead Strategies of Constraint Satisfaction.	CO4	6
B) Explain Constraint Satisfaction with an example?	CO4	6
C) Explain Travelling salesman problem in AI?	CO4	6

*** End ***